

Exercises: Generalized Linear Model

Valentina Zangirolami - valentina.zangirolami@unimib.it

December 21, 2023

(Referring to the theoretical parts: 20, 21, 22, 23, 24, 25, 26)

1 Exercise 1

To meet competition or cope with economic slowdowns, corporations sometimes undertake a “reduction in force” (RIF), in which substantial numbers of employees are terminated. Federal and various state laws require that employees be treated equally regardless of their age. In particular, employees over the age of 40 years are in a “protected” class, and many allegations of discrimination focus on comparing employees over 40 with their younger coworkers. Here are the data for a recent RIF:

Terminated	Over40: No	Over40: Yes
Yes	17	71
No	564	835

Exercise 1.1

(a) Choose an appropriate response variable and then a regression model. Justify your answer.

(b) Further, find the estimated regression model.

BINARY OUTPUT SO PROBIT OR LOGIT

$$\eta_i = \beta_1 + \beta_2 D_i; \quad D_i = \begin{cases} 1 & \text{OVER 40} \\ 0 & \text{BELOW 40} \end{cases}$$

$$\text{LOG}\left(\frac{\pi_i}{1-\pi_i}\right) = \eta(\pi_i) = \eta_i$$

$$Y_i \sim \text{BINARY}(\pi_i)$$

• RESPONSE VARIABLE: TERMINATED
• LOGISTIC REGRESSION

WE NEED TO FIND THE ESTIMATED REGRESSION MODEL $\rightarrow$ FIND $\hat{\beta}_1$ AND $\hat{\beta}_2$

$$\pi_i = \frac{e^{\beta_1 + \beta_2 D_i}}{1 + e^{\beta_1 + \beta_2 D_i}}$$

$$P(Y_i = 1 | D_i = 0) = \frac{e^{\beta_1}}{1 + e^{\beta_1}}$$

$$P(Y_i = 0 | D_i = 0) = 1 - \frac{e^{\beta_1}}{1 + e^{\beta_1}} = \frac{1}{1 + e^{\beta_1}}$$

$$P(Y_i = 1 | D_i = 1) = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}}$$

$$P(Y_i = 0 | D_i = 1) = 1 - \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} = \frac{1}{1 + e^{\beta_1 + \beta_2}}$$

$$\frac{P(Y_i = 1 | D_i = 1)}{P(Y_i = 0 | D_i = 1)} = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} \cdot \frac{1 + e^{\beta_1}}{1} = e^{\beta_1 + \beta_2} \rightarrow e^{\hat{\beta}_1 + \hat{\beta}_2} = \frac{71}{835} \rightarrow \hat{\beta}_2 = 1.035$$

$$\frac{P(Y_i = 1 | D_i = 0)}{P(Y_i = 0 | D_i = 0)} = \frac{e^{\beta_1}}{1 + e^{\beta_1}} \cdot \frac{1 + e^{\beta_1}}{1} = e^{\beta_1} \rightarrow e^{\hat{\beta}_1} = \frac{17}{564} \rightarrow \hat{\beta}_1 = -3.50$$

$$\eta_i = -3.5 + 1.030 D_i$$

Exercise 1.2

Software gives the estimated slope $\hat{\beta}_2 = 1.0371$ and its standard error $SE(\hat{\beta}_2) = 0.2755$. Transform the results to the odds scale. Summarize the results (using confidence intervals) and write a short conclusion.

$$\hat{\beta}_2 = 1.0371 \quad SE(\hat{\beta}_2) = 0.2755$$

$$\frac{P(y=1|D=1)}{P(y=1|D=0)} = \frac{e^{\beta_1 + \beta_2}}{e^{\beta_1}} = e^{\beta_2} = e^{1.0371} = 2.81$$

THE ~~ODDS~~ OF OLDER THAN 40 TO BE TERMINATED ARE EQUAL TO THE ~~ODDS~~ OF YOUNGER THAN 40 TO BE TERMINATED MULTIPLIED BY 2.81

$$CI = \left(e^{\hat{\beta}_2 - SE(\hat{\beta}_2) \cdot Z_{1-\alpha/2}}, e^{\hat{\beta}_2 + SE(\hat{\beta}_2) \cdot Z_{1-\alpha/2}} \right)$$

$1-\alpha = 0.99; \alpha = 0.01; \alpha/2 = 0.005; 1-\alpha/2 = 0.995$

$$= \left(e^{1.0371 - 0.2755 \cdot 1.96}, e^{1.0371 + 0.2755 \cdot 1.96} \right)$$

$$= (1.61, 4.33)$$

Exercise 1.3

If additional explanatory variables were available, for example, a performance evaluation, how would you use this information to study the RIF?

- ADD TO THE MODEL

$$y_i = \beta_0 + \beta_1 D_i + \beta_2 x_i$$

- OR DO EVALUATION TO SEE

IF IT IS USEFUL

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

2 Exercise 2

The acquisition literature suggests that takeovers occur either due to conflicts between managers and shareholders or to create a new entity that exceeds the sum of its previously separate components. Other research has offered managerial hubris as a third option, but it has not been studied empirically. Recently, some researchers revisited acquisitions over a 10-year period in the Australian financial system. A measure of CEO overconfidence was based on the CEO's level of media exposure, and a measure of dominance was based on the CEO's remuneration relative to the firm's total assets. They then used logistic regression to see whether CEO overconfidence and dominance were positively related to the probability of at least one acquisition in a year. To help isolate the effects of CEO hubris, the model included explanatory variables of firm characteristics and other potentially important factors in the decision to acquire. The following table summarizes the results for the two key explanatory variables:

Covariates	Estimates	SE
Overconfidence	0.0878	0.0402
Dominance	1.5067	0.0057

Exercise 2.1

Write the estimated regression model and interpret the coefficients estimates.

OUTPUT: BINARY $\begin{cases} 1 \text{ AT LEAST 1 ACQUISITION} \\ 0 \text{ NO ACQUISITION} \end{cases}$

SO WE PROBIT OR LOGIT. I CHOOSE LOGIT

- $M_i = \beta_1 + \beta_2 x_{i1} + \beta_3 x_{i2}$
- $\text{LOG} \left(\frac{\pi_i}{1-\pi_i} \right) = M_i = \eta(\pi_i)$
- $Y_i \sim \text{BERNOULLI}(\pi_i)$

$$\text{LOG} \left(\frac{\pi_i}{1-\pi_i} \right) = \hat{\beta}_1 + 0.0878x_{i1} + 1.5067x_{i2}$$

INTERPRETATION:

- WHEN KEEPING DOMINANCE CONSTANT, IF WE INCREASE OVERCONFIDENCE BY 1 UNIT, ON AVERAGE, WE INCREASE THE LOG ODDS OF ACQUISITION BY A FACTOR OF 0.0878
- WHEN KEEPING OVERCONFIDENCE CONSTANT, IF WE INCREASE DOMINANCE BY 1 UNIT, ON AVERAGE, WE INCREASE THE LOG ODDS OF ACQUISITION BY A FACTOR OF 1.5067

Exercise 2.2

Perform the significance tests and determine whether the variables are significant at the 0.05 level.

SIGNIFICANCE TEST OF OVERCONFIDENCE

$$\frac{\ln\left(\frac{\pi_1}{1-\pi_1}\right)}{\ln\left(\frac{\pi_0}{1-\pi_0}\right)} = \frac{\beta_1 + \beta_2(x_{1i} + 1) + \beta_3 x_{2i}}{\beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i}} \rightarrow \frac{\frac{\pi_1}{1-\pi_1}}{\frac{\pi_0}{1-\pi_0}} = \frac{e^{\beta_1 + \beta_2(x_{1i} + 1) + \beta_3}}{e^{\beta_1 + \beta_2 x_{1i} + \beta_3}} = e^{\beta_2}$$

$$\alpha = 0.05$$

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad Z_{obs}^2 \stackrel{H_0}{\sim} N(0,1) \quad Z_{obs}^2 = \frac{\hat{\beta}_2 - \beta_2}{0.0402} = 2.184$$

$$\alpha^{obs} = P(Z_2 | Z_{obs}^2) = 2(1 - \Phi(|Z_{obs}^2|)) \approx 0.02$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.05$ WE REJECT H_0 , SO THE COVARIATE "OVERCONFIDENCE" IS SIGNIFICANT

SIGNIFICANCE TEST OF DOMINANCE

$$\alpha = 0.05$$

$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$$

$$Z_{obs}^3 \stackrel{H_0}{\sim} N(0,1) \quad Z_{obs}^3 = \frac{\hat{\beta}_3 - \beta_3}{0.0057} = 264.3$$

$$\alpha^{obs} = P(Z_3 | Z_{obs}^3) = 2(1 - \Phi(|Z_{obs}^3|)) \approx 0$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.05$ WE REJECT H_0 , SO THE COVARIATE DOMINANCE IS SIGNIFICANT

Exercise 2.3

Estimate the odds ratio for each variable and construct a 95% confidence interval.

$$1-\alpha = 0.95; \alpha = 0.05; \alpha/2 = 0.025; 1-\alpha/2 = 0.975$$

OVERCONFIDENCE

$$\frac{\log\left(\frac{\pi_1}{1-\pi_1}\right)}{\log\left(\frac{\pi_0}{1-\pi_0}\right)} = \frac{\beta_1 + \beta_2(x_{2i}+1) + \beta_3x_{2i}}{\beta_1 + \beta_2x_{2i} + \beta_3x_{2i}} \rightarrow \frac{\frac{\pi_1}{1-\pi_1}}{\frac{\pi_0}{1-\pi_0}} = \frac{e^{\beta_1 + \beta_2(x_{2i}+1) + \beta_3x_{2i}}}{e^{\beta_1 + \beta_2x_{2i} + \beta_3x_{2i}}} = e^{\hat{\beta}_2}$$

AN INCREASE IN 1 UNIT OF OVERCONFIDENCE LEADS TO AN INCREASE OF $e^{\hat{\beta}_2}$ IN THE ODDS

$$CI = \left(e^{\hat{\beta}_2 - SE(\hat{\beta}_2)Z_{1-\alpha/2}}, e^{\hat{\beta}_2 + SE(\hat{\beta}_2)Z_{1-\alpha/2}} \right)$$

DOMINANCE

$$\frac{\log\left(\frac{\pi_1}{1-\pi_1}\right)}{\log\left(\frac{\pi_0}{1-\pi_0}\right)} = \frac{\beta_1 + \beta_2x_{2i} + \beta_3(x_{2i}+1)}{\beta_1 + \beta_2x_{2i} + \beta_3x_{2i}} \rightarrow \frac{e^{\beta_1 + \beta_2x_{2i} + \beta_3(x_{2i}+1)}}{e^{\beta_1 + \beta_2x_{2i} + \beta_3x_{2i}}} = e^{\beta_3}$$

AN INCREASE OF ONE UNIT IN DOMINANCE LEADS TO AN INCREASE OF e^{β_3} IN THE ODDS OF DOMINANCE

$$CI = \left(e^{\hat{\beta}_3 - SE(\hat{\beta}_3)Z_{1-\alpha/2}}, e^{\hat{\beta}_3 + SE(\hat{\beta}_3)Z_{1-\alpha/2}} \right)$$

3 Exercise 3

Let consider a dataset on the number of research articles published by 915 graduate students in biochemistry PhD programs. The variables for this dataframe are

- **art**: count of articles produced during last 3 years of PhD
- **fem**: factor indicating gender of student, with levels Men and Women
- **mar**: factor indicating marital status of student, with levels Single and Married
- **kid5**: number of children aged 5 or younger
- **phd**: prestige of PhD department
- **ment**: count of articles produced by PhD mentor during last 3 years

Exercise 3.1

Choose an appropriate response variable and then a regression model. Justify your answer.

- $n = 915$
- **RESPONSE VARIABLE: ART** → # ARTICLES PRODUCED DURING 3 YEARS of PhD
- **OUTPUT IS NOT BINARY** → Poisson
- $Y_i \sim \text{Poisson}(\mu_i)$
- $\eta_i = \mu_i$
 - ↑ 1 MEN
↑ 0 WOMEN
 - ↑ 1 SINGLE
↑ 0 MARRIED
 - ↑ 0 KID5
 - ↑ PRESTIGE PhD
 - ↑ # ARTICLES MENTOR
- $\eta_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \beta_6 X_{6i}$

Exercise 3.2

Complete the following table.

Coefficients	Estimates	SE	Z-obs	P-value
β_1	0.304617	0.102981	2.958	0.0031
β_2	-0.224594	0.054613	(1)	(2)
β_3	0.155243	(4)	(3)	0.0114
β_4	(5)	0.040127	-4.607	(6)
β_5	(8)	0.026397	(7)	0.6271
β_6	0.025543	0.002006	12.733	< 2e-16

Then, interpret the coefficient β_6 .

(1) $\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases} \quad Z_{obs}^2 \sim N(0,1) \quad Z_{obs}^2 = \frac{\hat{\beta}_1 - \beta_2}{0.054613} = -4.11$

(2) $\alpha_{obs}^2 = P(|Z_2| > |Z_{obs}^2|) = 2(1 - \Phi(|Z_{obs}^2|)) = 3.92 \cdot 10^{-3}$

(3) $\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases}$

$\alpha_{obs}^3 = 0.0149 \quad \alpha_{obs}^3 = 2(1 - \Phi(|Z_3^{obs}|)) \rightarrow \Phi(|Z_3^{obs}|) = 1 - \frac{\alpha_{obs}^3}{2} = 2.53519$

(4) $SE(\hat{\beta}_3) = \frac{\hat{\beta}_3 - \beta_3}{Z_3^{obs}} = \frac{0.155243 - 0}{2.53519} = 0.0613$

(5) $\begin{cases} H_0: \beta_4 = 0 \\ H_1: \beta_4 \neq 0 \end{cases} \quad Z_4^{obs} \sim N(0,1)$

$Z_4^{obs} = \frac{\hat{\beta}_4 - \beta_4}{SE(\hat{\beta}_4)} \rightarrow \hat{\beta}_4 = Z_4^{obs} \cdot SE(\hat{\beta}_4) = -0.18$

(6) $\alpha_{obs}^4 = P(|Z_4| > |Z_{obs}^4|) = 2(1 - \Phi(|Z_{obs}^4|)) = 4.0852 \cdot 10^{-6}$

(7) $Z_5^{obs} \sim N(0,1)$

$\begin{cases} H_0: \beta_5 = 0 \\ H_1: \beta_5 \neq 0 \end{cases} \quad \alpha_{obs}^5 = 2(1 - \Phi(|Z_5^{obs}|)) \rightarrow \Phi(|Z_5^{obs}|) = 1 - \frac{\alpha_{obs}^5}{2} = 0.9758$

(8) $\hat{\beta}_5 = SE(\hat{\beta}_5) \cdot Z_5^{obs} = 0.0123$

INTERPRETATION β_6

WHEN KEEPING ALL THE COVARIATES CONSTANT, IF WE INCREASE THE NUMBER OF ARTICLES PRODUCED BY PhD MEMBERS DURING LAST 3 YEARS BY ONE UNIT, ON AVERAGE, WE INCREASE THE ODDS OF COUNT ARTICLES BY PND OF 0.025543

Exercise 3.3

Knowing that the null deviance is equal to 1817.4 while the residual deviance is equal to 1634.4. Perform a statistical test about the overall significance. Specify the hypothesis, the test statistic and the p-value.

$$D(NVCC) = 1817.4$$

$$D(MOD.) = 1634.4$$

$$\begin{cases} H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \\ H_1: \neg H_0 \end{cases}$$

$$W_{obs} \sim \chi^2_{p-p_0}$$

↑
6-1=5

$$W_{obs} = \text{res}\left(\frac{\hat{\ell}(MOD)}{\hat{\ell}(NVCC)}\right) = 2 \left\{ \hat{\ell}(MOD) - \hat{\ell}(NVCC) + \hat{\ell}(SAT) - \hat{\ell}(SAT) \right\}$$

↓ ⊖ ↓ ⊖ ↓ ⊖

$$= D(NVCC) - D(MODEL) = 1817.4 - 1634.4 = 183$$

$$p^{obs} = P(W \geq W_{obs}) = 1 - P(W \leq W_{obs}) \approx 0$$

SINCE $p^{obs} < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ I REJECT H_0 AT ALL CONFIDENCE LEVEL

Exercise 3.4

Knowing the value of the following quantity

$$\sum_{i=1}^n y_i \log(\hat{\mu}_i) - \hat{\mu}_i = -642.0261$$

Find the log-likelihood of the saturated model. Further, interpret the results in terms of goodness of fit.

Poisson $f(\mathbf{y};) = \frac{e^{-\sum \mathbf{y}_i} \prod \mathbf{y}_i!}{\prod \mathbf{y}_i!}$

- $\hat{l}(\text{Mod}) = \sum -\mathbf{y}_i + \mathbf{y}_i \log(\mathbf{y}_i)$
- $\tilde{l}(\text{Mod}) = n[\bar{y} + \bar{y} \log(\bar{y})]$
- $\tilde{l}(\text{SAT}) = \sum \mathbf{y}_i + \mathbf{y}_i \log(\mathbf{y}_i)$

$\hat{l}(\text{MODEL}) = -642.0261$

$D(\text{MODEL}) = 1817.4$

$D(\text{RES.}) = 1634.4$

$D(\text{RES}) = 2 \{ \tilde{l}(\text{SAT}) - \hat{l}(\text{MODEL}) \} \rightarrow \tilde{l}(\text{SAT}) = \frac{D(\text{RES})}{2} + \hat{l}(\text{MODEL}) = 175.1239$

Good Model? If $D(\text{Mod})$ is in $[0, n-p] \rightarrow$ since $n-p < 1634.4$ Model not good ;)

4 Exercise 4

A researcher is interested in how variables, such as **GRE** (Graduate Record Exam scores), **GPA** (grade point average) and prestige of the undergraduate institution (**Rank**), effect admission into graduate school. The response variable (**Admit**), admit/don't admit, is a binary variable. **Rank** takes on the values 1 through 4. The total number of observations is 400.

Exercise 4.1

Write the equation of the regression model using probit. Which other kind of model can we use?

- $n = 400$
- BINARY OUTCOME I USE LOGIT or **PROBIT**
 - $\begin{cases} 1 & \text{RANK} = 2 \\ 0 & \text{OTHERWISE} \end{cases}$
 - $\begin{cases} 1 & \text{RANK} = 3 \\ 0 & \text{OTHERWISE} \end{cases}$
 - $\begin{cases} 1 & \text{RANK} = 4 \\ 0 & \text{OTHERWISE} \end{cases}$
- $\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 D_{5i}$
 - β_1 GRE
 - β_2 GPA
- $Y_i \sim \text{BERNOULLI}(\pi_i)$
- $\text{LOG}\left(\frac{\pi_i}{1-\pi_i}\right) = \eta_i = \eta(\pi_i)$

$$\text{PROBIT} \cdot \eta(\pi_i) = \eta_i = \Phi(\pi_i)^{-1}$$

$$Y_i \sim \text{BERNOULLI}(\pi_i)$$

Exercise 4.2

Knowing the estimates of regression coefficients using a probit model, such that

Coefficients	Estimates
β_1	-2.38684
β_2	0.00138
β_3	0.47773
β_4	-0.41540
β_5	-0.81214
β_6	-0.93590

Interpret them.

. β_2 : KEEPING EVERYTHING CONSTANT, IF I INCREASE GRE BY ONE UNIT I INCREASE THE PROB. OF ADMISSION BY 0.00138

~ " "

, " "

Exercise 4.3

Now, let consider another link function (should have already been specified in ex. 4.1).
Write the regression model.

LOGIT

- $Y_i \sim \text{BERNOULLI}(\pi_i)$

- $\text{log}\left(\frac{\pi_i}{1-\pi_i}\right) = \eta(\pi_i) = \eta_i$

- $\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 D_{3i} + \beta_4 D_{4i} + \beta_5 D_{5i}$

$\begin{cases} 1 & \text{SRE} \\ 0 & \text{OTHERWISE} \end{cases}$

$\begin{cases} 1 & \text{SPA} \\ 0 & \text{OTHERWISE} \end{cases}$

$\begin{cases} 1 & \text{RANK} = 1 \\ 0 & \text{OTHERWISE} \end{cases}$

$\begin{cases} 1 & \text{RANK} = 2 \\ 0 & \text{OTHERWISE} \end{cases}$

$\begin{cases} 1 & \text{RANK} = 3 \\ 0 & \text{OTHERWISE} \end{cases}$

Exercise 4.4

We are interested in performing a test for comparing models. Let consider the previous model as full model, and the restricted model as a model which just includes **Rank**. Knowing the value of the observed test statistic, $w^{obs} = 16.449$, specify the hypothesis, the theoretical test statistic and p-value. Discuss about the result.

• FULL MODEL: $M_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \beta_4 D_{3i} + \beta_5 D_{4i} + \beta_6 D_{5i}$

• RESTRICTED $M_i = \beta_1 + \beta_4 D_{3i} + \beta_5 D_{4i} + \beta_6 D_{5i}$

$w_{obs} = 16.449$

$\begin{cases} H_0: \beta_2 = \beta_3 = 0 \\ H_1: \neg H_0 \end{cases}$

$w_{obs} \sim \chi^2_{p-p_0}$

$6-4=2$

$\alpha^{obs} = P(W_3 \geq w_{obs}) = 1 - P(W < w_{obs})$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$!
REJECT H_0 AT ALL CONFIDENCE LEVEL

Exercise 4.5

Knowing the deviance of the previous restricted model $D(\text{restricted}) = 458.52$, and the deviance of the null model $D(\text{null}) = 499.98$, perform a test about overall significance by specifying the hypothesis, the test statistic and the p-value.

$$D(\text{REST}) = 458.52$$

$$D(\text{NULL}) = 499.98$$

$$\begin{aligned} \text{find } D(\text{Model}) &\rightarrow \overset{\text{p-value}}{\downarrow} W^{\text{OBS}} = D(\text{REST}) - D(\text{Model}) \\ &\downarrow \\ D(\text{Model}) &= D(\text{REST}) - W^{\text{OBS}} \\ &= 458.52 - 16.449 = 442.071 \end{aligned}$$

TEST OVERALL SIGNIFICANCE

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0 \\ H_1: \text{THO} \end{cases}$$

$$W^{\text{OBS}} \sim \chi^2_{p-p_0} \\ \uparrow \\ 6-1, 5$$

$$\begin{aligned} W^{\text{OBS}} &= \log \left(\frac{\hat{\ell}(\text{Model})}{\hat{\ell}(\text{NULL})} \right) = 2 \left\{ \hat{\ell}(\text{Model}) - \hat{\ell}(\text{NULL}) - \hat{\ell}(\text{SAT}) + \hat{\ell}(\text{SAT}) \right\} \\ &= D(\text{NULL}) - D(\text{Model}) = 499.98 - 442.071 \\ &= 57.909 \end{aligned}$$

$$\alpha^{\text{OBS}} : P(W \geq W^{\text{OBS}}) = 1 - P(W \leq W^{\text{OBS}}) \sim 0$$

SINCE $\alpha^{\text{OBS}} < \alpha$ FOR $\alpha = 0.1$; $\alpha = 0.05$; $\alpha = 0.01$ REJECT
H₀ AT ALL CONFIDENCE LEVELS

Exercise 4.6

Referring to the full model, we know

- $\hat{\beta}_3 = 0.804038$
- $z^{obs} = 2.423$

Interpret the value of β_3 . Perform the test of significance and discuss about the result.

$$M_i = \beta_1 + \beta_2 X_{1i} + \beta_3 X_{2i} + \beta_4 D_{3i} + \beta_5 D_{4i} + \beta_6 D_{5i}$$

GRE GPA RANK = 2
1 0 OTHERWISE RANK = 3
1 0 OTHERWISE RANK = 4
1 0 OTHERWISE

WHEN KEEPING EVERYTHING CONSTANT, ON AVERAGE,
IF I INCREASE GPA BY ONE UNIT WE INCREASE
THE LOG-ODDS AT A FACTOR OF 0.804038

$$\begin{cases} H_0: \beta_3 = 0 \\ H_1: \beta_3 \neq 0 \end{cases} \quad Z_{obs}^3 \overset{H_0}{\sim} N(0,1) \quad Z_{obs}^3 = 2.423$$

$$\alpha^{obs} = P(|Z_3| \geq |Z_{obs}^3|) = 2(1 - P(Z_3^3 \leq |Z_{obs}^3|)) = 0.15$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1$ I REJECT H_0 AT THIS CONFIDENCE LEVEL
FOR $\alpha = 0.01$ $\alpha^{obs} > \alpha$ SO I DO NOT REJECT H_0

Exercise 4.7

Now, we would like to evaluate our models in terms of mis-classification. Let consider two logistic regression models with probit and logit link functions, involving all covariates. The confusion matrix for each model corresponds to

Confusion Matrix and Statistics

Prediction	Reference	
	0	1
0	157	40
1	116	87

Figure 1: Probit

Confusion Matrix and Statistics

Prediction	Reference	
	0	1
0	157	40
1	116	87

Figure 2: Logit

Compute the mis-classification error and interpret the results.

NOT IN EXAM

5 Exercise 5

A researcher reports an experiment on the toxicity to the tobacco budworm *Heliothis virescens* of doses of the pyrethroid *trans*-cypermethrin to which the moths were beginning to show resistance. Batches of 20 moths of each sex were exposed for three days to the pyrethroid and the number in each batch that were dead or knocked down was recorded. The results were

Dose	
Sex	1 2 4 8 16 32
Male	1 4 9 13 18 20
Female	0 2 6 10 12 16

Exercise 5.1

Write an appropriate regression model using $\log_2(Dose)$ and *Sex*. Justify your answer.

$$\begin{aligned}
 & \bullet Dose \\
 & \bullet Sex \quad \begin{cases} 1 & \text{if Male} \\ 0 & \text{otherwise} \end{cases} \\
 & \quad \swarrow \log_2(Dose) \\
 & \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_1 + \beta_2 x_{1i} + \beta_3 D_{1i}
 \end{aligned}$$

Exercise 5.2

Considering the following table

Covariates	Estimates	SE
Intercept	-3.4732	0.4685
Dose	1.0642	0.1311
Sex	1.1007	0.3558

Interpret the estimated regression coefficients and obtain a confidence interval at 95% confidence level.

$$1 - \alpha = 0.95 \rightarrow \alpha = 0.05 \rightarrow \alpha/2 = 0.025 \rightarrow 1 - \alpha/2 = 0.975$$

- WHEN KEEPING EVERYTHING CONSTANT, IF I INCREASE THE DOSE BY ONE UNIT, ON AVERAGE I HAVE AN INCREASE IN THE LOG ODDS OF BEING DEATH OF 1.0642

$$CI = (\hat{\beta}_3 - SE(\hat{\beta}_3) Z_{1-\alpha/2}; \hat{\beta}_3 + SE(\hat{\beta}_3) Z_{1-\alpha/2})$$